

University of British Columbia
 Department of Electrical & Computer Engineering
 MECH 464/563 EECE 442/589: Introduction to Robotics

Assignment #4

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Read Chapter 3 from my notes (ch3_velkin.pdf), Corke, Sec. 8.1.4, 8.2.1, 8.2.2 (Jacobian singularities).

Find the manipulator geometric Jacobian for the robots shown in Figures 1, 2 and 3.

For each case, find the robot singular configurations.

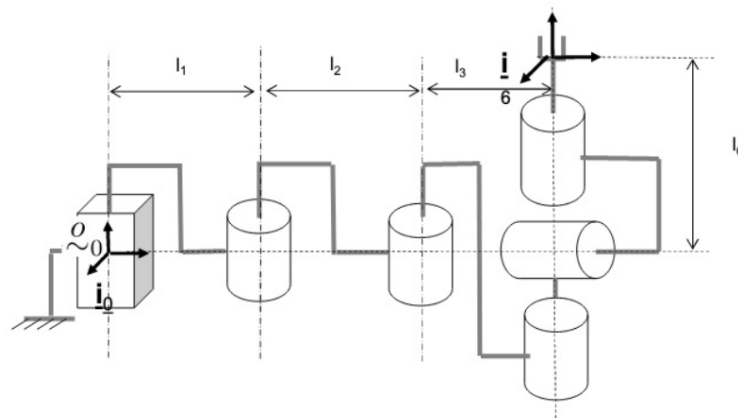


Figure 1: SCARA robot with vertical axis in the base.

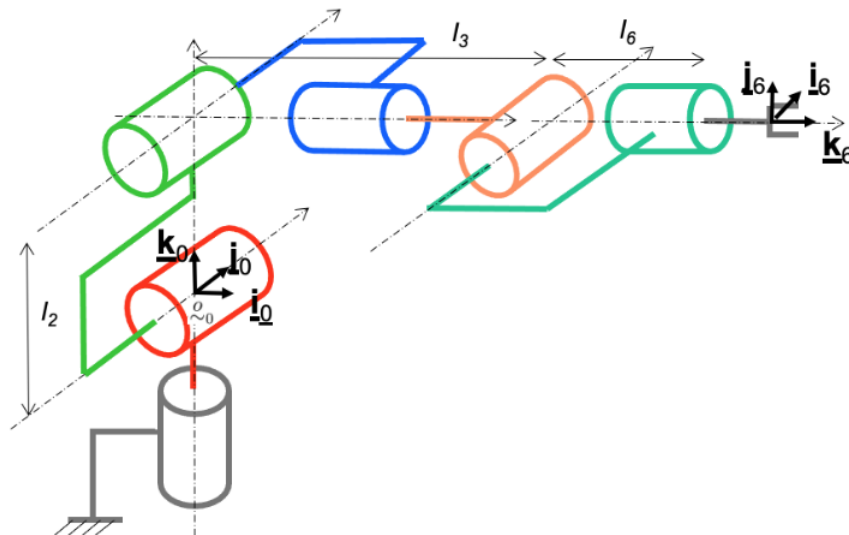


Figure 2: Anthropomorphic (PUMA without offsets).

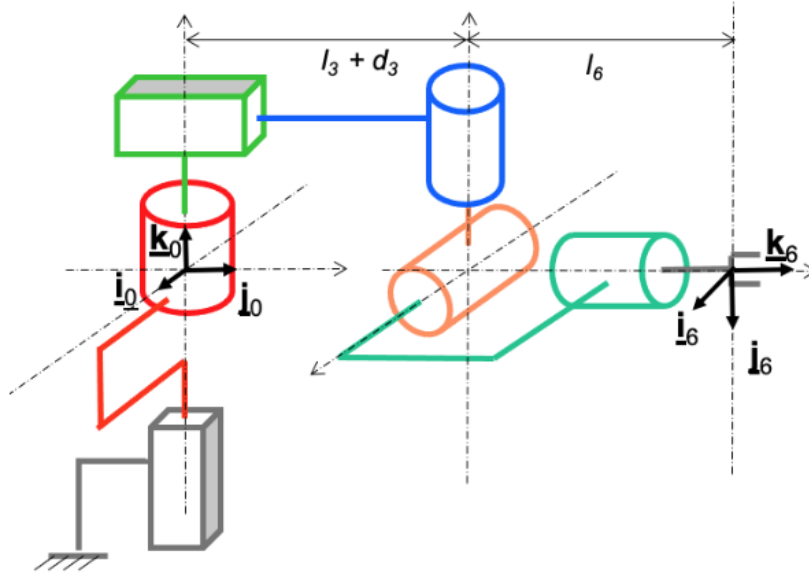


Figure 3: Cylindrical robot.

1 Figure 1: SCARA

Given that the first joint is prismatic and the rest are revolute, we have that the geometric Jacobian is

$$J = \begin{bmatrix} k_0 & k_1 \times (o_6 - o_1) & k_2 \times (o_6 - o_2) & k_3 \times (o_6 - o_3) & k_4 \times (o_6 - o_4) & k_5 \times (o_6 - o_5) \\ 0 & k_1 & k_2 & k_3 & k_4 & k_5 \end{bmatrix}.$$

However, by the DH convention, we would have that the co-ordinate origins for o_3 , o_4 , and o_5 are coincident, because the axes of those three joints pass through each other. Also, we have that k_0, k_1, k_2 are the same unit vector, so we simplify, to get the geometric jacobian:

$$J = \begin{bmatrix} k_0 & k_0 \times (o_6 - o_1) & k_0 \times (o_6 - o_2) & k_3 \times (o_6 - o_3) & k_4 \times (o_6 - o_3) & k_5 \times (o_6 - o_3) \\ 0 & k_0 & k_0 & k_3 & k_4 & k_5 \end{bmatrix}.$$

Finally, to expose the singular configurations, we multiply (on the left) by a matrix with determinant one, which will not change the singularity of the resulting product:

$$\begin{bmatrix} I_{3 \times 3} & (o_6 - o_3) \times \\ 0 & I_{3 \times 3} \end{bmatrix} J = \begin{bmatrix} k_0 & k_0 \times (o_3 - o_1) & k_0 \times (o_3 - o_2) & 0 & 0 & 0 \\ 0 & k_0 & k_0 & k_3 & k_4 & k_5 \end{bmatrix}.$$

We analyze the singularity conditions:

- Analyzing the diagram, we can see that $k_4 \perp k_3$ and $k_4 \perp k_5$ at all times. However, we have the possibility that $k_3 \parallel k_5$, in which case the matrix is singular. This corresponds to a gimbal lock in the spherical wrist.
- We have by default that $k_0 \perp k_0 \times (o_3 - o_{1,2})$, so for the velocity components to cause a non-singularity, we would require $(o_3 - o_1) \parallel (o_3 - o_2)$. That happens, for instance, when both joints 2 and 3 are fully extended.

2 Figure 2: PUMA without offsets

We have that all the joints are revolute, giving us a Jacobian of the form:

$$J = \begin{bmatrix} k_0 \times (o_6 - o_0) & k_1 \times (o_6 - o_1) & k_2 \times (o_6 - o_2) & k_3 \times (o_6 - o_3) & k_4 \times (o_6 - o_4) & k_5 \times (o_6 - o_5) \\ k_0 & k_1 & k_2 & k_3 & k_4 & k_5 \end{bmatrix}.$$

However, if we follow the DH convention we have that $k_1 = k_2$, $o_0 = o_1$ and $o_3 = o_4 = o_5$, so we can simplify:

$$J = \begin{bmatrix} k_0 \times (o_6 - o_0) & k_1 \times (o_6 - o_0) & k_1 \times (o_6 - o_2) & k_3 \times (o_6 - o_3) & k_4 \times (o_6 - o_3) & k_5 \times (o_6 - o_3) \\ k_0 & k_1 & k_1 & k_3 & k_4 & k_5 \end{bmatrix}.$$

Finally, to find the singularities, we use the fact that we have a spherical wrist by premultiplying with a matrix of determinant 1:

$$\begin{bmatrix} I_{3 \times 3} & (o_6 - o_3) \times \\ 0 & I_{3 \times 3} \end{bmatrix} J = \begin{bmatrix} k_0 \times (o_3 - o_0) & k_1 \times (o_3 - o_0) & k_1 \times (o_3 - o_2) & 0 & 0 & 0 \\ k_0 & k_1 & k_1 & k_3 & k_4 & k_5 \end{bmatrix}.$$

Then, we observe the singularities:

- Like last time, we observe from the diagram that $k_3 \perp k_4$ and $k_4 \perp k_5$, but we have the possibility, again, of gimbal lock, when $k_3 \parallel k_5$, causing a singularity.
- If $k_0 \parallel (o_3 - o_0)$, then $k_0 \times (o_3 - o_0) = 0$, and we have a singularity. This happens when the spherical wrist origin $o_3 = o_4 = o_5$ than axis 1.
- If $(o_3 - o_0) \parallel (o_3 - o_2)$, then we also have a singularity because then we have two linearly dependent columns.

3 Figure 3: Cylindrical robot

The first and third joints are prismatic and the rest are revolute, with a spherical wrist. With the DH co-ordinates, we can choose $o_0 = o_1 = o_2$ to be at the co-ordinate frame o_0 . Then, we have the geometric jacobian:

$$J = \begin{bmatrix} k_0 & k_1 \times (o_6 - o_1) & k_2 & k_3 \times (o_6 - o_3) & k_4 \times (o_6 - o_4) & k_5 \times (o_6 - o_5) \\ 0 & k_1 & 0 & k_3 & k_4 & k_5 \end{bmatrix}.$$

For the singularities, we use the the spherical wrist:

$$J = \begin{bmatrix} k_0 & k_1 \times (o_3 - o_1) & k_2 & 0 & 0 & 0 \\ 0 & k_1 & 0 & k_3 & k_4 & k_5 \end{bmatrix}.$$

- $k_3 \parallel k_5$ gives us a singularity as a result of a gimbal lock in the spherical wrist.
- Two prismatic joints in the shoulder gives us no other singularities.